1Agile Product Portfolio
Software Engineering Group Coursework

Student Second-hand Marketplace

Group number: #45

Student ID	Student Name
2944152	Yu-Ching Chang
xxxxx	xxxxx xxxxx
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1. Planning (20%)

Conceptual model
Personas □ Product Vision □ User Stories □ Features □ Prototype □ Tests

1.1 Personas

Persona Name	Role/User type	Goals	Pain Points	Related User Stories (IDs)
Mike	Graduating Student / Seller	Wants to sell textbooks and furniture quickly before leaving the UK.	Thinks eBay is too complicated and Facebook Marketplace has too many scammers or non-students.	S1 S2
Andy	The Fresher / Buyer	Wants to buy cheap, affordable items (kettles, textbooks) for the dorm.	New items are too expensive; second-hand shops are too far from campus.	S1
Tommy	The Seller	Wants to edit owned products details and change password and username	Hard to locate Owned Products to edit detail, and struggling to find the security page for regular password update.	S2
Tim	The Fresher	Wants to search for certain items and utilize a price range that fits my needs.	Hard to find items that match the fresher‘s specific budget and save my time, using names to search items.	S3

Note: You need to have at least two personas. The personas should be used to justify the product vision and user stories.

1.2 Product Vision

FOR University students and staff
WHO want to buy affordable items or sell used goods to reduce waste
The **Student Second-hand Marketplace** is a web-based campus second-hand trading platform
THAT allows users to easily list, discover, and exchange second-hand items within the campus

UNLIKE general platforms like Facebook Marketplace or eBay
OUR PRODUCT is tailored for the university community, focuses on face-to-face campus exchange, and reducing campus waste.

Note: This product vision should be derived from the defined personas and guides backlog prioritisation and design decisions.

1.3 Product Backlog and User Stories

Story ID	Feature (epics)	Story Title	User Story	Acceptance Criteria	Priority (Must / Should / Could)	Justification of prioritisation /Trade-off	Story Owner
S1	Item Management	Post an Item	User Story As a Seller I want to post an item with price and description, so that other students can see what I am selling.	**Given** I am on the "Sell Item" page, **When** I enter the item name, price, and description and click "Post", **Then** the item appears in the marketplace list immediately.	Must	Core feature. Without displaying items, buyers cannot buy products.	xxxxx xxxxx xxxxx xxxxx
S2	User Profile	Edit Information	User Story As a Seller, I want to edit and show the information of my product, so that buyers can know how to contact me and know the products I am selling.	**Given** I am on the "User Profile" page **And** I have a product named "whale" listed for sale **When** I click the "edit item"button for the product I am selling **And** I modify the product information and click "Post" **Then** the modified product appears in the marketplace **And** buyers can see my contact information and the updated product details	Must	Core feature Without the correct information of items, buyers cannot buy products fairly.	Yu-Ching Chang

S3	Search Bar	Search items	User Story As a Buyer I want to buy certain items from the website that match my budget and my requirements.	**Given** I am on the "Home page" **When** I type 'whale' into the search bar **And** I select the price range "£20" to "£50" next to the search bar **And** I click the search button **Then** the market should display all items that match my all conditions.	Must	Core feature Without search bar, buyers cannot find the items they want efficiently	xxxxx xxxxx xxxxx xxxxx

1.4 Sprint 1 Summary

Item	**Sprint 1 (Weeks 2-6)** Discovery, MVP design and partial implementation	**Sprint 2(Weeks 7-11)** Refinement, testing & evaluation
Planned stories (Plan - P)	S1	**S2, S3**
Delivered stories (Done - D)	S1	**S2, S3**
Stories not completed (No - N)		
Key change planned for Sprint 2	1. Data Persistence: Transition from basic JSON handling to a more robust storage structure. 2. Authentication: Implement User Login/Logout logic to support S2 (User Profile). 3. UI/UX Enhancement: Improve the CSS styling of the marketplace based on initial feedback.	**N/A**

Note: Stories marked "P" or "N" should be reflected upon in Sprint learning or retrospectives.

2. Design (20%)

2.1 Design Models and Evidence

Diagram	**Type (Class / Sequence)**	**Purpose**	**Related User Stories (Story IDs)**	**Design decision rationale**	**Reference**

D1	Class	Domain Structure	S1, S2	Clarifies data structure and relationships between User and Product models based on the database schema.	README.md / classDiagram_code20260213.pdf
D2	Use Case	System Functionality	S1, S2, S3	Visualizes interactions between Guests, Registered Members, and the market system boundary.	README.md / sprint2diagram/UseCaseDiagram.jpeg
D3	Sequence	Logic of "Post Item"	S1	Models the end-to-end interaction flow from the web form submission to JSON file persistence.	Refer to: README.md / sprint2diagram/Sequence Diagram2.1.pdf
D4	Sequence	User Profile Management	S2	Illustrates session verification and the process of updating user information while maintaining data consistency.	README.md
D5	Sequence	Search & Classification	S3	Details how product data is loaded across categories and filtered based on user-defined keywords and price ranges.	README.md

Note: You are **not penalised** for modifying or discarding models if justified by learning.

2.2 Stories and Evidence

For each story marked as "Done" in Sprint 1 must be mapped to at least one test.

Story ID	Evidence Type (UML Diagram / Git Commit ID(s) / Test Evidence: testcases / their results)	Reference (file / commit / screenshot)	Acceptance Criteria Met (Yes / Partially / No)	Notes (if partially completed)

	Class Diagram	70e828a	Partially	Improved diagrams in 5ad7dea, 5cb7244
	Class Diagram	5ad7dea, 5cb7244	Yes	
	Use case Diagram	d1da62ff	Yes	
S1	Diagram git commit	f37e3c0 (Sequence Diagram)	Yes	
S1	Git commit	9232634 (completed upload function and list all details of items)	Yes	
S1	Test Evidence: test cases / their results	BUS_Manual test.pdf Git commit: a49a885	Yes	
S2	Diagram git commit	5231eab (Sequence Diagram)	Yes	
S2	Git commit	Cdcaf58, a1c6833 (personal dashboard, improve features) 1653372, c7512e8 (change	Yes	

		password/username)		
S2	Test Evidence: test cases / their results	BUS_Manual test.pdf Git commit: a49a885	Yes	
S3	Git commit	41dbb70c b8bbf0cf	Yes	
S3	Diagram git commit	53e2170e ae9bd559	Yes	
S3	Test Evidence: test cases / their results	BUS_Manual test.pdf Git commit: a49a885	Yes	

Note: If a story is marked "Done" without evidence, it **will not be treated as complete**.

3. Working Prototype and Implemented Features (20%)

Submit a **short video (3-5 minutes)** that demonstrates the implementation of 2 core end-to-end features + refinement of Sprint 1 feature. Each demonstrated feature must be traceable to user stories and acceptance criteria. The video should be submitted under **the "Coursework video"** assignment on Canvas.

4. Testing and Evaluation (20%)

Story ID	Acceptance Criteria	Test Performed	Outcome (Pass / Partial / Fail)	Accessibility Checked	Notes (If partially tested)
S1	Items are saved to the database and displayed in the marketplace immediately.	Manual functional test of "Post Item" form submission and JSON data persistence.	Pass	Keyboard only	Detailed test logs and screenshots refer to: BUS_Manual test.pdf.

S2	Profile information is updated and modified products reflect changes across the market.	Manual validation of Profile dashboard updates and product edit logic.	Pass	Keyboard only	Detailed evidence and update logic refer to: BUS_Manual test.pdf.
S3	The system displays items matching both keyword query and selected price range filters.	Manual search testing using 'whale' keyword and varied price range categories.	Pass	Keyboard only	Detailed test scenarios and filter results refer to: Test Scenarios_20260213.pdf.

Note: You may have more than one test per story. At least one test per story is required.

APPENDIX

A. Group Contribution

Story ID	Story Title	Predicted workload in hours	Actual workload in hours	Status (TODO/In progress / DONE)	Owner name
S1	Post an Item	15	18	DONE	xxxxx xxxxx xxxxx xxxxx
S2	User Profile	12	10	DONE	Yu-Ching Chang
S3	Search Item	10	12	DONE	xxxxx xxxxx xxxxx xxxxx

B. Individual Contribution

Team member	Role (s)	Stories Owned	Evidence (Git / Diagram / Test)	What could have been improved with more time?
Yu-Ching Chang	Personas, Product vision, S1 Backlog and User stories, Use case Diagram, S1 Sequence Diagram, S1 QA, S2 Developer, Agile Product Portfolio Document management	Story 2	Personas, Product vision, User stories, Agile Product Portfolio Document management: xxxxx xxxxx xxxxx Use case Diagram: d1da62f S1 Sequence Diagram: F37e3c0 S2 Coding: cdcaf58, a1c6833 (personal dashboard, improve features) 1653372, c7512e8 (change password/username)	Refactor the codebase to improve logic and coding style, and transition from server-side rendering to using RESTful APIs for better maintainability.

xxxxx xxxxx	Personas, Product vision, S1 Backlog and User stories, S1 Class Diagram, S1 QA, S2 Developer	Story 1	Personas, Product vision, Class Diagram, Sequence Diagram, Test Scenario, Unit testing, Integration	Data Visualization Coding
xxxxx xxxxx	S1 Code Developer, Debugging S2 Sequence Diagram, Use Case Diagram S3 Code Developer, Debugging Demo Video, Meeting logs record	Story 3	0102d9a8: "Register" Update 530b2e05: "Login" Create b8bbf0cf: "Edit" Create	Cleaner Code
xxxxx xxxxx	Personas, S1 UI/UX Designer , S2 & S3 testing, S2 Sequence Diagram, Use Case Diagram, Meeting logs record, Demo Video, Video Editing	Story 1	A49a885: "Test case" 6b87943: "UI/UX Enhancement"	UI/UX Enhancement, Testing
xxxxx xxxxx	S1 Developer debug S2 Sequence Diagram Use Case Diagram S3 developer DemoVideo	Story 3	Sequence diagram:5231eabc Bugfix:bb5d0169 S3 feature basic:41dbb70c	Cleaner code

C. Meeting Logs

You are required to schedule at least 2 meetings with your Customer (TA).

Meeting ID	**Team attendees**	**Customer (TA's name)**	**When**	**Agenda items**	**Actions and Owner**
M1	Yu-Ching Chang xxxxx xxxxx xxxxx xxxxx xxxxx xxxxx xxxxx xxxxx	xxxxx xxxxx	11/2/2026 10:40 - 11:00		Action: Ask for the suggestions about the whole projects Owner: All team members
M2	Yu-Ching Chang xxxxx xxxxx xxxxx xxxxx xxxxx xxxxx xxxxx xxxxx	xxxxx xxxxx	12/2/2026 14:00 - 14:30	1. Proposed 5 topics 2. Selected "Second-hand Marketplace" 3. Confirmed Sprint 1 Scope	Action: Start developing the first feature. Owner: All team members
M3	Yu-Ching Chang xxxxx xxxxx xxxxx xxxxx xxxxx xxxxx	xxxxx xxxxx	18/2/2026 10:40 - 11:00	1. Demo user story 1	Action: Demonstrate the results of sprint 1.

	xxxxx xxxxx				Owner: All team members
M4	Yu-Ching Chang xxxxx xxxxx xxxxx xxxxx xxxxx xxxxx xxxxx xxxxx	xxxxx xxxxx	18/3/2026 13:00 - 13:20	1. Demo user story 2	Action: Developing the rest of the features.. Owner: All team members
M5	Yu-Ching Chang xxxxx xxxxx xxxxx xxxxx xxxxx xxxxx xxxxx xxxxx	xxxxx xxxxx	25/3/2026 13:00 - 13:20	1. Demo user story 3	Action: Report the results of the whole project. Owner: All team members

D. Group Meeting

Meeting ID	**Team attendees**	**When**	**Agenda items**	**Actions and Owner**
M1	Yu-Ching Chang xxxxx xxxxx xxxxx xxxxx xxxxx xxxxx xxxxx xxxxx	13/2/2026 14:30 - 15:00	1. Initialize Git repository and project structure. 2. Decide on data persistence: Using JSON files for the MVP. 3. Define the directory structure for "seller" and "uploads".	**Action**: Set up the Flask environment and create basic routes. **Owner**: All team members
M2	Yu-Ching Chang xxxxx xxxxx xxxxx xxxxx xxxxx xxxxx xxxxx xxxxx	10/3/2026 14:00 - 15:00	1. Sprint 1 Retrospective: Review S1 "Post Item" results. 2. Discuss S2 (User Profile) logic: Session management and JSON linking. 3. Assign S3 (Search & Filter) development tasks.	**Action**: Implement Search functionality and category filtering. **Owner**: All team members

M4	Yu-Ching Chang xxxxx xxxxx xxxxx xxxxx xxxxx xxxxx xxxxx xxxxx	27/3/2026 14:00 - 15:00	1. Final UI/UX review: Standardize CSS styles and primary colors. 2. Debugging session: Fix flash message bugs and password field types. 3. Plan the 3-5 minute Demo Video recording.	**Action**: Refine UI styling and complete the Final Portfolio PDF. **Owner**: All team members

Note: Meeting logs are not marked directly and are used only as supporting evidence.